



EARN TO FIRE

Answer STEM questions. Fire the arena. Battle the bots.



Think

Your team answers robot science, maths and coding questions on an iPad.



Fire

Get one right and a real arena hazard fires: the flipper, the pit or the spinners.



Fight

Your drivers battle bots in the arena. Keep yours out of the pit!

Two teams. Two bots each. Two minutes.

Your group splits into **Red** and **Blue**. Two of each team drive the bots; the rest are the brains on the iPad. Your score is **STEM points + bot points**, so the brains and the drivers both win it for you.

What you'll learn

 **Forces & levers:** how a flipper lifts a robot

 **Code & logic:** loops, binary, IF statements

 **Circuits & electricity:** volts, amps, Ohm's law

 **Sensors & robot brains:** how robots sense the world

 **Gears & motion:** speed, RPM, gear ratios

 **Boss challenge:** energy, momentum and more

Got one wrong? The iPad explains the right answer, so you learn something either way.

How does the arena work?

Each iPad talks over WiFi to a small server. The server sends the command to a **Raspberry Pi**, a credit-card-sized computer, which switches the arena's motors, solenoids and servos. The flipper is a **solenoid**: a coil that becomes an electromagnet and kicks it up. **Look through the clear front of the arena base to see the tech!**

⚠ Safety: the arena hazards only move when our referee has **armed** it, and they switch themselves off after a few seconds. Never reach into the arena. Bots go in and out only when the referee says so.



SPLIT THE GROUP

"You're two teams: **Red** and **Blue**. In each team, **two of you drive a bot**. Everyone else is the **brains** on your team's iPad."

THE BRAINS

"Tap a weapon and answer the question. **Get it right and that weapon fires for real**, and you score its points. Get it wrong and the iPad tells you why. Then try a different weapon."

THE DRIVERS

"Keep your bots **out of the pit**. You score a point every second each bot survives. Push the other team's bots in and your team gets **50 points** for each one!"

TEAMWORK TIP

"Brains: watch where their bots are. **Fire the hazards when their bots are close!**"

SAFETY

"**Hands stay out of the arena**. If I say **STOP**, everyone stops."

GO

"Two minutes on the clock. Highest score wins. **Ready? Go!**"

Group of 10: ● **Red** 5 (2 drivers + 3 brains) vs □ **Blue** 5 (2 drivers + 3 brains). Odd numbers: the extra joins the brains. Small group? One driver per team can drive both bots in turn.



Tap a weapon on the iPad and answer its question.

Right answer = it FIRES and you score.

WEAPON	QUESTION TOPIC	POINTS	RECHARGE
 Flipper	Forces & levers	10	8 s
 The Pit	Circuits & electricity	15	20 s
 Spinner 1	Gears & motion	20	15 s
 Spinner 2	Code & logic	20	15 s
 Spinner 3	Sensors & robot brains	20	15 s
 MEGA SPIN	Boss challenge (all three spinners!)	50	30 s

**Bot points**

+1 every second, for each of your bots still in

+50 for each of the other team's bots pitted

Remember

Wrong answer? Read why, then wait 15 s before that weapon again. Try another meanwhile!

Recharge is shared. Fire it first and the other team has to wait.

YOUR SCORE = STEM POINTS + BOT POINTS

The 7-minute slot

0:00 Reset New round (everything). Split Red / Blue	0:30 Brief Read page 2. Drivers take controllers	1:30 Go Hands out. ARM. START	3:30 Stop DISARM. Read the Standings	4:00 Wrap Leaderboard. Teacher sheet	5:00 Next Bots back. Next group in
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About 50 groups a day. Longer slot? Play a second round and swap drivers and brains.

Start of each day

- ☐ Team iPads open `192.168.0.102/kids`, pick Red / Blue
- ☐ Referee iPad opens `/kids/admin.html` (PIN from the lead)
- ☐ Referee shows **SAFE**; iPads show **PRACTICE MODE**
- ☐ Answer one per weapon: scores move, arena doesn't
- ☐ Arena clear: **ARM**, fire each weapon, watch it stop
- ☐ **ALL STOP** works and returns to SAFE
- ☐ Bot batteries charged, spares charging

During the round

Tap **PITTED** under a bot the moment it drops. Mistap? **Undo**. Both of a team's bots pitted = **KNOCKOUT**: the round ends.

SAFE / ARMED	SAFE = answers score, nothing moves
START / STOP	The clock. STOP pauses questions
RESET	Clock back to 2:00, fresh fight
±5 / ±10	Fix STEM points by hand
Clear cooldowns	Unstick the weapons

⚠ Safety rules

- **DISARM before any hand goes near the arena.**
- **ALL STOP** = disarm, stop clock, all spinners off. Use it freely.
- Spinners run 8 s (MEGA 10 s) in a random direction, then stop; the pit comes back up after 10 s.
- Server drops out mid-spin? **Switch off arena power.**
- After any restart the arena comes back **SAFE**.

If something goes wrong

iPad: Cannot reach the arena	Check its WiFi, reload the page
Right answer, nothing moved	Is it ARMED? Look for <i>Pi FAILED</i> in the activity log
Referee iPad back at PIN screen	Server restarted. Log in again; it will be SAFE
5 wrong PINs	Locked for 60 s. Wait, then retry

Top team score today

Best STEM score today

[illegible]

Red / Blue = team totals from the referee iPad's Standings (STEM + bot points). **KO** = tick if both of one team's bots were pitted.

Your class played **Earn to Fire** with InfoSec Battle Bots at STEMfest Newcastle. In two teams, students answered STEM questions to fire real hazards in a robot combat arena while classmates drove the robots. Every wrong answer came with an explanation.

Topics covered (KS3)

Forces & levers	Fulcrums, Newton's laws, weight = mass $\times$ g, centre of gravity
Circuits & electricity	Ohm's law ($V = I \times R$), $P = V \times I$, series circuits, fuses, solenoids
Gears & motion	Speed = distance / time, RPM, gear ratios, circumference
Code & logic	Variables, loops, IF statements, binary, AND / NOT
Sensors & robot brains	Ultrasonic ranging, inputs vs outputs, actuators, microcontrollers

Back in class: five follow-up questions

1. A spinner weapon spins **twice as fast**. Why does it hit **four times** harder? ($KE = \frac{1}{2}mv^2$)
2. A 12 V battery drives a 4 Ω motor. What current flows, and how much power does it use?
3. Design a pit-avoiding robot: which sensor would you use, and what would your **IF** statement say?
4. Why do battle robots use a **low centre of gravity**? Sketch a robot that is hard to flip.
5. Count to 31 in binary on one hand. How high can you count with two hands?

Answers: (1) energy grows with speed squared, so $2^2 = 4$; (2) 3 A, 36 W; (3) e.g. a downward light or ultrasonic sensor: IF floor not detected THEN reverse; (4) it must be lifted further before it tips; (5) 1023 (10 bits).

How the arena works

The iPads talk over WiFi to a Windows server running a Node.js program. It checks each answer, then sends commands to a **Raspberry Pi**, which switches the arena's motors, solenoids and servos. It is real networking, coding and electronics: a good talking point for careers in engineering and cyber security.

Want more? Termly schools competitions

We're building towards a schools competition every term, working with local **STEM Ambassadors**. Our schools work is led by **Lewis Terry**. Get in touch to take part:

Email	info@infosecbattlebots.com (subject "Schools")
Schools page	www.infosecbattlebots.com/schools
Follow us	@InfosecBots on X
The game's code	github.com/Ben-PyroGuy-Doch/IBB-Arena-STEM-Challenge (open source)